

1. Objective

The purpose of this analysis is to understand player behaviour, satisfaction and preference by tracking and identifying key factors like total revenue, customer reviews, player engagement by different games and genres.

This analysis is significant because by analysing three datasets (games_description, games_revenue, games_reviews) which contain key information like game name, publisher, genre, number of purchases, unit prices, number of reviews from customers, number of English reviews and engagement tags, it can provide insights to help improving game design, maximizing revenue potential and offering players high quality captivating games from player behaviour and satisfaction.

2. Joining Method

The joining methods we used for this analysis were Inner Joins, and the primary key used to join the tables was Game_Id.

By using the Inner Join, we were able to join the games description, games review and games revenue tables to complete our analysis.

Games_description table was joined using Inner join to games revenue and games_reviews tables to link game details such as game_name, game_id, number of purchases, unit price, number_of_reviews_from_purchased_people, number_of_english_reviews, and helpful and funny tags.

We chose to use Inner joins as an Inner join can join two or more tables and will return data that is common to the joined tables which allows us to only use relevant and accurate data for our analysis.

3. Insight Explanation

Total Revenue

In this analysis, total revenue (Number_of_Purchases * Unit_Price) is used to analyse data to identify which game, genre, year and publisher generate the most income in the company

game_name	Total_Revenue
Counter	407366540
PUBG: B	105578436
Dota 2	91050736
Grand T	48013200
Terrari	41601392

For game, it shows Counter generates the most income to the company as Counter has the highest total revenue compared to other games, followed by PUBG: B as the second highest and Dota 2 as the third highest.

genre	TotalRevenue
Action	1258756620
Simulation	184076708
Role-Playing	73065268

For genre, Action generates the most income than other genres to the company as it has the highest total revenue, followed by Simulation as runner up and Role-Playing as second runner up. In addition, the table above also shows Action has a far more higher total revenue than the aggregate of Simulation and Role-Playing and shows around nine times more total revenue than Simulation, this indicates Action is a more popular genre in games than other genres which can produce more income to the company.

year_released	Total_Revenue
2012	427990780

For year, 2012 is the year that the company has the highest total revenue, which shows the company has the best sales of games compared to other years.

publisher	Total_Revenue
Valve	604943693
KRAFTON	105578436
Rocksta	57446151
Electro	49495729
CD PROJ	47518459

For publisher, Valve is the best publisher as it has the highest total revenue and shows domination in total revenue than other publisher as Valve has around six times more total revenue than the runner up publisher (KRAFTON), this indicates Valve may have a better reputation than other publisher and the games published by Valve results to gain more income to the company.

game_name	genre	Total_Revenue	Ranking
Counter	Action	407366540	1
Baldur'	Adventure	8738780	1
Stardew	Role-Playing	16423105	1
Garry's	Simulation	33098650	1
Fishing	Sports & Racing	2136744	1
Bloons	Strategy	9372222	1

Using Window Function, it shows a deeper view to see which game has the highest total revenue in each genre.

For Action, Counter has the highest total revenue with \$407366540 than the other games with Action genre.

For Adventure, Baldur' has the highest total revenue with \$8738780 than the other games with Adventure genre.

For Role-Playing, Stardew has the highest total revenue with \$16423105 than the other games with Role-Playing genre.

For Simulation, Garry's has the highest total revenue with \$33098650 than the other games with Simulation genre.

For Sports & Racing, Fishing has the highest total revenue with \$2136744 than the other games with Sports & Racing genre.

For Strategy, Bloons has the highest total revenue with \$9372222 than the other games with Strategy genre.

Reviews

Reviews is another key metric used in this analysis to identify player satisfaction and engagement, which helps to provide future feedback like game design improvement to the company. Helpful and Funny tags are some key factors used to identify player engagements.

total_customer_reviews	total_purchase	Review_Percentage
47603561	74572039	63.84%

There are 63.84% of customers left reviews out of total purchase in all games, indicates more than half of customers like to leave reviews for a game in their purchase to provide feedback of the game.

total_games	games_with_eng_reviews_major	English_Reivew_Percentage
235	170	72.34

In the reviews, there are 72.34% of games having English as the majority in the total reviews, which means there are many games have more than half of the total reviews that are written in English and this shows English is the major language used by players to leave comments than other language.

game_name	Helpful	number_of_reviews_from_purchased_people	Helpful_Percentage
Terrari	3926660	1061260	370.00
All In	0	5	0.00

For Helpful Engagement Ratios, Terraria is the game that has a highest percentage of Helpful tags to total reviews, which is around 370%. All In is the game that has a lowest percentage of Helpful tags, which is 0%.

genre	total_helpful_in_each_genre	total_reviews_in_each_genre	Helpful_Percentage
Action	25914462	34734548	74.61
Sports & Racing	93274	515140	18.11

Action is the genre that has the highest percentage of Helpful tags to total reviews, which is 74.61%. Sports & Racing has the lowest percentage, which is 18.11%. It also shows a pattern of genre having a larger number of reviews will have a higher Helpful tags percentage.

game_name	Funny	number_of_reviews_from_purchased_people	Funny_Percentage
Tom Cla	1121920	1121920	100.00
Team Fo	1118530	1118530	100.00
Garry's	945676	945676	100.00

For Funny Engagement Ratios, there are a lot of games having the same highest and lowest percentage. Tom Cla is the game which has the highest Funny percentage with the most number of Funny tags and number of reviews from purchased customers than other games which holds the same Funny percentage.

game_name	Funny	number_of_reviews_from_purchased_people	Funny_Percentage
Terrari	530630	1061260	50.00
Apex Le	474409	948818	50.00
Rust	456664	913328	50.00

On the other hand, Terraria is the game has the lowest Funny percentage with the most number of Funny tags and number of reviews from purchased customers than other games which holds the same Funny percentage, which is 50%.

genre	total_funny_in_each_genre	total_reviews_in_each_genre	Funny_Percentage
Role-Playing	2504568	2961231	84.58
Adventure	1417908	2328712	60.89

Role-Playing is the genre that has the highest percentage of Funny tags to total reviews, which is 84.58%. Adventure has the lowest percentage, which is 60.89%.

game_name	genre	Total_Reviews	Review_Rank
Counter	Action	8313603	1
Baldur'	Adventure	582585	1
Stardew	Role-Playing	644043	1
Garry's	Simulation	945676	1
Need fo	Sports & Racing	96628	1
Bloons	Strategy	289266	1

Using Window Function, it shows a deeper view to see which game has the highest total reviews in each genre.

For Action, Counter has the highest total reviews with 8313603 than the other games with Action genre.

For Adventure, Baldur' has the highest total reviews with 582585 than the other games with Adventure genre.

For Role-Playing, Stardew has the highest total reviews with 644043 than the other games with Role-Playing genre.

For Simulation, Garry's has the highest total reviews with 945676 than the other games with Simulation genre.

For Sports & Racing, Need for Speed has the highest total reviews with 96628 than the other games with Sports & Racing genre.

For Strategy, Bloons has the highest total reviews with 289266 than the other games with Strategy genre.

4. Conclusion

For this project, we analysed data on game sales, reviews, and publisher data to identify which games, genres and publishers generated the most revenue and engagement.

We joined tables using Inner joins to connect the games_description, games_reviews, and games_revenue tables using the primary key Game_Id. This allowed us to bring together information such as sales, review counts, and engagement data for analysis. We also used the window function Rank, to rank games by their total revenue and reviews within each genre, this function allowed us to identify top-performing games and trends.

Upon reviewing this data, we found that many top performing games and genres contribute massively to the company's revenue with Counter-Strike being the top contributing game earning over \$407 million and the Action genre making over \$1 Billion in revenue.

2012 was the greatest year in sales for the company earning over \$420 Million.

We can see from the data analysed that a major portion of reviews that were left are in English, which suggests that many of the players can speak English.

Valve is a major publisher bringing in over \$604 million in Revenue, making it one of the top contributors to the companies' overall earnings.

After analysing the engagement data, we can see that Terraria received the highest number of Helpful tags with over 1 million reviews and 100% tagged as helpful. All in All had the lowest amount with only 5 reviews and 0% tagged as helpful.

Action is the genre that had the highest Helpful tags percentage with 74.61% and Sports & Racing had the lowest percentage with 18.11%.

The game with the most number of funny tags was Tom Cla with over 1.1 million reviews and 100% funny tags and the lowest was Terrari which had over 1 million reviews but only 50% tagged funny.

The genre with the highest funny tags was role-playing at 84.57% while the lowest was adventure with 60.89 %.

Recommendations

We recommend the company to focus on the high performing games and genres, particularly those in Action and Role-Playing as those produce the most revenue and engagement. Games such as Terrari, Apex le and Rust that have low tag percentages could benefit from some targeted marketing campaigns to help boost engagement or game upgrades for player satisfaction.

The company should consider creating a new series for the top selling games such as Counter and Dota 2, these popular games have the potential to entice old players to purchase the new series boosting revenue. By using the popular game titles, it reduces the risk of creating an entirely new title that may not be successful.

The company should continue to publish English language games with Valve as they are highly reputable and generate a large portion of their revenue.

The company should frequently analyze sales data, review tags and publisher data to recognize any trends and to track performance that will help make informed business decisions.

Limitations

This analysis is based on historical data therefore may not be up to date and accurate.